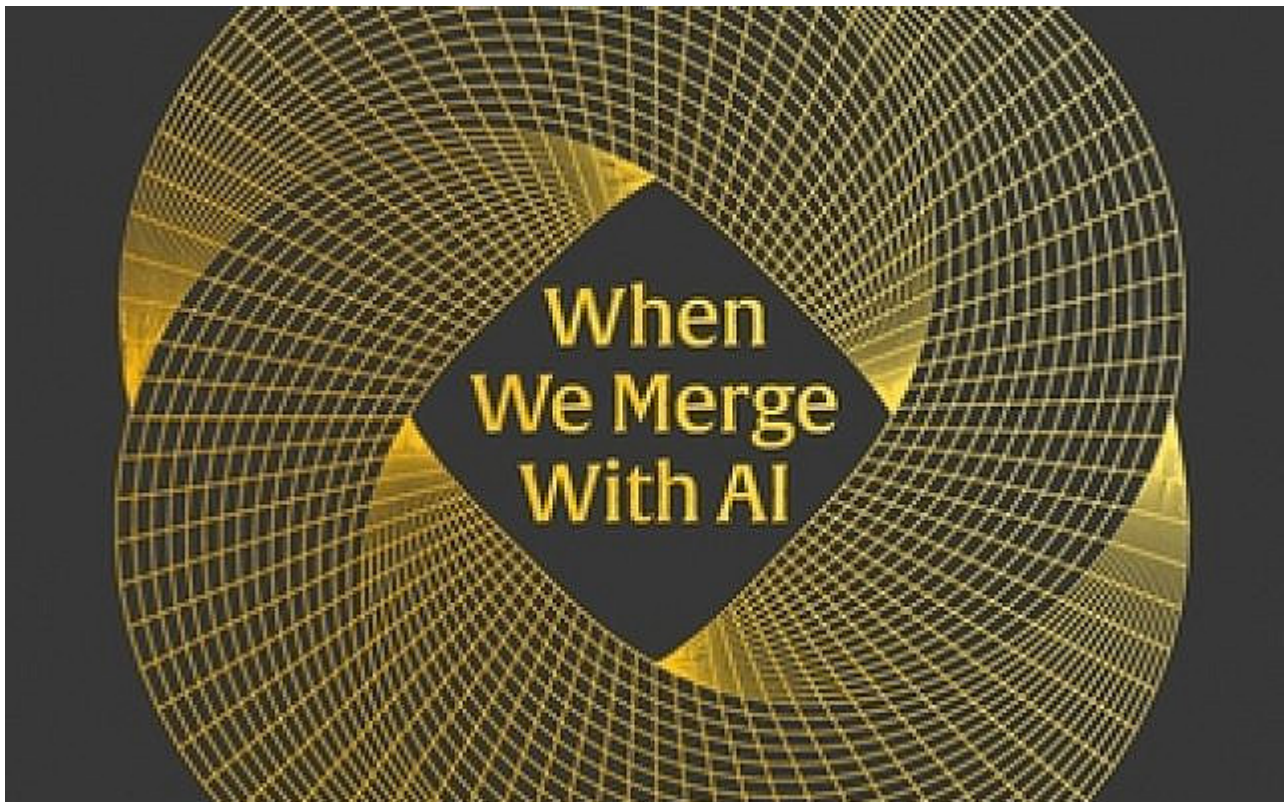




THE BLOGS

Daniel S. Smith

The Singularity is Nearer



Review of Ray Kurzweil's [forthcoming](#) (June 2024) book *The Singularity is Nearer: When We Merge with AI*. Penguin Publishing Group.

Renowned futurist Ray Kurzweil envisions a future where those under

eighty and in good health have the potential to live forever. He predicts that by the 2030s, we will be able to extend the neocortex of our brains into the cloud, enabling a massive increase in human intelligence. Kurzweil's latest work, *The Singularity is Nearer*, takes readers on a journey from ignorance to enlightenment, shedding light on the incredible possibilities that await us. Even if you've previously overlooked Kurzweil's predictions over the past four decades, now is the time to take notice. His track record of accurate forecasts demonstrates that we can indeed predict the future, and his insights into what lies ahead are invaluable.

Many who would have brushed Kurzweil aside as a heretic in 2005 when he published the *Singularity is Near*, reviving John the Baptist's prediction, "the kingdom of heaven is near" (Matthew 3:2), in 1990 the *Age of Intelligent Machines*, or 1999 *Age of Spiritual Machines*, are more likely to take his arguments seriously in 2024. Incredible advancements in technology over the past few decades, particularly in the fields of AI and biotech, have lent significant credibility to Kurzweil's predictions. Microsoft co-founder Bill Gates describes Kurzweil as, "the best person I know at predicting the future of artificial intelligence."

Kurzweil is intervening into a century of literature and debate, right up to spring of 2023. He has been working in the field of artificial intelligence, a term he does not like because it makes it seem "less real," for over sixty years. The book serves as a historiography of machine intelligence and the myriad debates therein.

In 1950 Alan Turing asked "Can machines think?" Pioneering computer scientist John Von Neumann made the first reference to the Singularity, writing a few years [after](#) Turing: "the ever-accelerating progress of technology" would yield "some essential singularity in the history of the race." In 1956, John McCarthy [defined AI](#) as "getting a computer to do things which, when done by people, are said to involve intelligence." In

1965, British Mathematician Irving John Good [predicted](#) an impending “intelligence explosion.” In that same year, Herbert Simon, a scientist who co-founded the field of artificial intelligence, [forecasted](#) by 1985, “machines will be capable of doing any work a man can do.” In 1993, Vernor Vinge wrote his seminal [essay](#): “The Coming Technological Singularity: How to Survive in the Post-Human Era,” arguing: “Within thirty years, we will have the technological means to create superhuman intelligence. Shortly after, the human era will be ended.”

In his 2005 book *The Singularity is Near*, Kurzweil defines the singularity as “an expansion of human intelligence by a factor of trillions through merger with its nonbiological form.” This will happen so rapidly that life will be “irreversibly transformed.” In *The Singularity is Nearer*, Kurzweil predicts in 2045 humanity will be, “Freed from the enclosure of our skulls, and processing on a substrate millions of times faster than biological tissue, our minds will be free to grow exponentially, ultimately expanding our intelligence millions-fold. This is the core of my definition of the Singularity.” The laws of physics, “allow for a continuation of exponential growth until non biological intelligence is trillions of times more powerful than all of human civilization today, contemporary computers included.” This intelligence will be too much for planet earth, and therefore engulf the entire universe.

Critics of Kurzweil such as Microsoft co-founder Paul Allen & Mark Greaves of Schmidt Futures [describe](#) his claims as premature. [Oren Etzioni](#), CEO of the [Allen Institute for Artificial Intelligence](#), is also doubtful about the imminence of superhuman AI: “Exponentials are very important. If we extrapolate exponentials, we can be exponentially wrong.” Mathematician Roger Penrose argued in his 1989 book *the Emperor’s New Mind* [that](#) some facets of human thinking can never be emulated by a machine. Philosopher John Searle has also [argued against](#) humanity achieving machine sapience, whereas engineer and the “godfather of

nanotechnology” Noam Chomsky thinks the singularity is science fiction. Philosopher Hubert Dreyfus [said that](#) AI is impossible in a 1965 RAND corporate memo entitled “Alchemy and Artificial Intelligence,” in which he concluded that the ultimate goals of AI research were as unachievable as were those of alchemy. The computer scientist Joseph Weizenbaum described the idea as obscene, anti-human and immoral. Pulitzer Prize winner Douglas Hofstadter [considered](#) it over-promising. Virtual-Reality (VR) pioneer Jaron Lanier [emphasizes](#) the importance of preserving individual creativity and personal expression in the digital age, warning against the homogenization of human experiences through technology.

Yet Kurzweil is doubling down again, arguing the rate of change is itself accelerating. He notes that today 39 percent of couples have met online. Who would have believed this in 2005?

How couples meet. Courtesy: Statista

In 2005, we were in the fourth epoch of technological development. According to Kurzweil, we are expected to pass the Turing Test by 2029, marking the transition to the fifth epoch. This prediction was first introduced in his 1999 book *The Age of Spiritual Machines*.

As we enter the 2030s, the fifth epoch will be characterized by a significant expansion of our cognitive abilities. This will be achieved by connecting the neocortex of our brain to the cloud, a concept Kurzweil explored in his 2012 book *How to Create a Mind*. For the sixth epoch, provided we are not limited by the speed of light, we can fill the entire universe with our intelligence by the year 2200. His predictions are based on his analysis of exponential growth in technological advancements.

In his 1990 book *The Age of Intelligent Machines*, Kurzweil predicted: “A

computer will defeat the human world chess champion around 1998, and we'll think less of chess as a result." He was one year off, as DeepBlue defeated chessmaster Garry Kasparov in 1997. In 2015, AlphaGo, an AI developed by Google's DeepMind, defeated the European Go champion Fan Hui. This victory marked the first time an AI had beaten a human professional Go player on a full-sized board without a handicap. With all of this progress, why would Kurzweil back down now?

Kurzweil argues AI will not be our competitor, but rather an extension of ourselves. The fifth epoch will involve brain-computer interfaces and will take "seconds to minutes (for us) to explore ideas unimaginable to present-day humans." This will benefit humankind, compared to life hundreds of years ago which was, "labor-intensive, poverty filled, and disease and disaster prone."

Life is getting exponentially better, yet we hardly notice because the news media tends to amplify tragedies as opposed to steady improvement. Constant fear mongering which plays toward our primal instincts leads to a more pessimistic view of society, for, "it's easier to share videos of disaster, but gradual progress doesn't generate dramatic footage.....This crowds out our capacity to assess positive developments that unfold slowly."

Kurzweil is a "technology optimist" who takes a historical exponential as opposed to an intuitive linear view of human progress. Linear growth is steady; exponential growth becomes explosive, for "we won't experience one hundred years of technological advance in the twenty-first century; we will witness on the order of twenty thousand years of progress." He claims Moore's Law has nothing to do with Intel and Thomas Moore, and has in fact been occurring since the 1880s, for, "It was the fifth, not the first, paradigm to bring exponential growth to the price/performance of computing."

His optimism set off a debate with Bill Joy of Sun Microsystems whose famous 2000 [Wired magazine essay](#), “Why the Future Doesn’t Need Us,” is more pessimistic. This is part of a larger divide with folks like Elon Musk and Stephen Hawking on the potential perils of artificial general intelligence (AGI). Ethicist and founder of the Machine Intelligence Research Institute (MIRI) Eliezer Yudkowsky argues the only way to deal with the threat of AGI is [“to shut it all down.”](#) Yudkowsky predicts, “If somebody builds a too-powerful AI, under present conditions, I expect that every single member of the human species and all biological life on Earth dies shortly thereafter.” He predicts a “hard takeoff” versus Robin Hanson’s “soft takeoff.” Kurzweil says he falls somewhere in the middle.

Citing Steven Pinker’s 2011 book *Better Angels of Our Nature* and 2018 book *Enlightenment Now*, as well as Peter Diamandis and Steven Kotler’s 2012 book *Abundance*, Kurzweil believes the state of the world keeps improving. He uses fifty graphs to show gradual progress over the past century, such as a decline in the rates of poverty, violence and child labor. He expects AI to accelerate these trends. Other optimists include OpenAI CEO Sam Altman who argues: “A.I. will be “the greatest force for economic empowerment and [a lot of people getting rich](#) we have ever seen.”

Courtesy: Cambridge University Press

In the technological pessimist's most extreme expression, Ted Kaczynski, the unabomber, called, violently, for an "anti-tech revolution." Kurzweil wrote in the *Age of Spiritual Machines*:

"Kaczynski is not talking about a contemplative visit to a nineteenth-century Walden Pond, but about the species dropping all of its technology and reverting to a simpler time. Although he makes a compelling case for the dangers and damages that have accompanied industrialization his proposed vision is neither compelling nor feasible. After all, there is too little nature left to return to, and there are too many human beings. For better or worse, we're stuck with technology."

Steven Pinker notes: "Pessimism can be a self-fulfilling prophecy," so it is best we accept the inevitable and make the most of it. Yuval Harari [writes](#), "In the twenty-first century, those who ride the train of progress will acquire divine abilities of creation and destruction, while those left behind will face extinction." Kurzweil says the nonbiological part of our intelligence "will combine the pattern-recognition powers of human

intelligence with the memory- and skill-sharing ability and memory accuracy of machines,” and thus will make it far more powerful than biological intelligence.

Kurzweil argued in *The Singularity is Near*: “any significant derailment of the overall advancement of technology is unlikely. Even epochal events such as two world wars (in which on the order of one hundred million people died), the cold war, and numerous economic, cultural, and social upheavals have failed to make the slightest dent in the pace of technology trends.” Over the past two centuries, technological advancements have created a positive feedback loop, leading to improvements in various aspects of human well-being, “Our merger with our technology has aspects of a slippery slope, but one that slides up toward greater promise, not down into Nietzsche’s abyss.” This will continue, as nanobots may [reverse pollution](#) from earlier industrialization.

For example, there has been a rise in the percentage of homes with electricity and computers, a proliferation in the availability of radios and televisions, an increase in life expectancy, and a rise in US GDP per capita. However, as Senator Robert F. Kennedy [famously stated](#), GDP measures everything “except that which is worthwhile,” suggesting that while these advancements have certainly improved certain aspects of human life, they may not necessarily reflect a holistic view of well-being.

Yuval Harari [notes](#) suicide has gone up in industrialized countries, “it is an ominous sign that despite higher prosperity, comfort and security, the rate of suicide in the developed world is also much higher than in traditional societies.” South Korea has rapidly industrialized since 1985, yet the suicide rate in that same period increased fourfold. Wealthy nations like Switzerland and Japan have more than twice as many suicides per capita than Peru and Ghana. Harari argues this may be because, “We don’t become satisfied by leading a peaceful and prosperous existence. Rather,

we become satisfied when reality matches our expectations. The bad news is that as conditions improve, expectations balloon.” Could life extension technologies could potentially help reduce the rates of suicide by giving people more hope for the future?

Some could be forgiven for wondering whether technological advancement has really benefited society. Do students with smartphones, tablets and computers learn better than if they only had a few books, a teacher, a notepad, and a pencil? What about the mental health problems posed by social media?

Writers like Adam Garfinkle, David Brooks and George Will are **concerned we have forgotten** how to “dwell with a text.” Yuval Harari does not own a smartphone, for he believes it is impossible to have perspective if you are constantly scrolling. He meditates for two hours a day, and takes month out of each year to go on a silent retreat with no electronics. Of course most of us are not so lucky and are forced to use these technologies, whereas Harari’s husband, Itzik Yahav, who Yuval describes as his “internet of all things,” manages his work. The increasing integration of technology into our lives **has been proven** to lessen empathy, and the drawbacks paired with the benefits are the paradox at the heart of the book, for one of Kurzweil’s principles is the respect for human consciousness.

As one indicator of progress, Kurzweil shows that democracy has spread rapidly around the world over the past century. Sure, the right to vote has been extended. But how much do our votes matter if the algorithm knows us better than we know ourselves, and can manipulate us, as the 2016 Cambridge Analytica Scandal showed? Brain scanners can now predict our actions and desires before we are aware of them. Yuval Harari notes: “What’s the point of having democratic elections when the algorithms know not only how each person is going to vote, but also the underlying

neurological reasons why one person votes Democrat while another votes Republican?” Harari [continues](#):

“Artificial intelligence and biotechnology might soon overhaul our societies and economies – and our bodies and minds too – but they are hardly a blip on the current political radar. Present-day democratic structures just cannot collect and process the relevant data fast enough, and most voters don’t understand biology and cybernetic well enough to form any pertinent opinions. Hence traditional democratic politics is losing control of events, and is failing to prevent us with meaningful visions of the future.”

Kurzweil doubts our political system will have evolved to answer these questions by the time AI passes the Turing Test, which is why we should push candidates to talk more about AI now so we are better able to manage it.

Kurzweil’s ultimate goal is to show the benefits outweigh the costs, urging: “Careful use of AI to provide openness and transparency while minimizing its potential to be used for authoritarian surveillance or to spread disinformation.” Combining his pattern recognition theory of mind (PRTM) with the LOAR will allow us to vastly extend our intelligence, and hopefully think of ways to avert the worst before it happens. This is quite the gamble, for he warns the same technologies that could empower us to cure cancer could be used by terrorists to unleash a deadly bioweapon.

A clear example of the benefits outweighing the costs are technological advancement for people with disabilities, who have seen vast improvements in their quality of life. As an inventor, Kurzweil’s advancements in speech recognition have led to the development of assistive technologies that help people with disabilities perform tasks that might otherwise be impossible, such as communicating, accessing

information, and controlling devices. Kurzweil has proposed the idea of using brain-computer interfaces (BCIs) to allow people with paralysis or other disabilities to control computers and other devices using their brainwaves. Life extension could lead to breakthroughs in treating diseases and conditions that disproportionately affect people with disabilities.

The author is not concerned about technological inequality. He cites smartphones as a case in point. At first, perhaps only the super-rich had access, but within years they became so cheap to mass-produce that now practically everybody has one. The same is true with vaccines. In his 2014 book *Superintelligence*, Oxford University philosopher Nick Bostrom [argues](#) “social elites” will gain first access to biological enhancement mechanisms and inspire a “culture shift” among everybody else: “Many of the initially reluctant might join the bandwagon in order to have a child that is not at a disadvantage relative to the enhanced children of their friends and colleagues.” A domino effect will ensure, assuming everybody can access these therapies.

Yuval Harari disagrees. He [writes](#) that in the 20th century medicine aimed to heal the sick, whereas in the twenty-first century medicine will increasingly aim to upgrade the healthy. There is hardly any reason to believe this will benefit the masses the same as elites, for

“The age of masses may be over, and with it the age of mass medicine. As human soldiers and workers give way to algorithms, at least some elites may conclude there is no point in providing improved or even standard levels of health for masses of useless poor people, and it is far more sensible to focus on upgrading a handful of superhumans beyond the norm.....Unlike in the twentieth century, when the elite had a state in fixing the problems of the poor because they were militarily and economically vital, in the twenty-first century the most efficient (albeit ruthless) strategy might be to let go of the useless third-class carriages,

and dash forward with the first class only.”

The concern is that the elites may find the populace superfluous given the rise of nonhuman intelligence, and therefore take the attitude of Marie Antionette and “let them eat cake.”

Harari’s opinion is worth urgently considering, for Kurzweil says we are “entering the steep part of the exponential.” [Eliezer Yudkowsky](#) argues in his 1996 book *Staring Into the Singularity*: “Don’t describe Life after Singularity in glowing terms. Don’t describe it at all.” But Kurzweil does not see the merger of humans and machines as something indescribable, but rather something that is already happening. Our intelligence is augmented exponentially by our constant access to smartphones, which is unprecedented because humans and machines are making decisions together.

In the Enlightenment, Rene Descartes said, *Cogito ergo sum*, or “I think, therefore I am.” Alan Turing helped set off the field of machine intelligence by asking “can machines think?” Yuval Harari [argues](#) intelligence is decoupling from consciousness, the difference being, “Intelligence is the ability to solve problems. Consciousness is the ability to feel things, such as pain, joy, love, and anger.”

In the 18th century, John Locke [wrote](#): “Since it is the understanding, that sets man above the rest of sensible beings, and gives him all the advantage and dominion, which he has over them; it is certainly a subject, even for its nobleness, worth our labour to inquire into.” John Searle argued consciousness could be infused into machines: “So the first step is to figure out how the brain does it and then build an artificial machine that has an equally effective mechanism for causing consciousness.” Kurzweil [believes](#): “In this view a dog is also conscious but somewhat less than a human. An ant has some level of consciousness, too, but much less that of

a dog. The ant colony, on the other hand, could be considered to have a higher level of consciousness than the individual ant; it is certainly more intelligent than a lone ant.” It matters whether or not machines are conscious, for it is on this basis that we can decide whether or not they should have rights.

Max Tegmark of the Future of Life Institute [defines](#) consciousness as subjective experience. The 2018 Cambridge Declaration of Consciousness concluded that consciousness is not exclusive to humans. In the future, it may be possible to transfer consciousness from our brains to computers. By augmenting the neocortex, we can enhance our subjective consciousness, experiencing the world in new ways. Kurzweil envisions, “We’ll be able to send nanobots into the brain noninvasively through the capillaries,” bypassing invasive procedures. This would mark the first significant neocortex revolution since the last one two million years ago, potentially enabling us to expand our intelligence a million-fold. In Kurzweil’s view, those who embrace this augmentation will far surpass those with unaugmented biological brains, leading to an unprecedented cognitive leap forward.

The good news is we will be able to back our brains up to the cloud, just like we do with our documents in Microsoft Office, so our experiences and records will be preserved regardless of what befalls our brain. We will also be able to download new skills in an instant. By the 2030s, we will be able to bring dead loved ones back using all of their data. A recent political attack-ad levied by the super-pac “The Lincoln Project” recreates US presidential candidate Donald Trump’s late father, Fred Trump, [disparaging his son](#). Who is to say which “replicants” can and cannot be created? By the early 2040s, us mere humans would not be able to tell the difference between our partner and a clone. Kurzweil collected all of his late father, Frederic Kurzweil’s writings and created a “Dad Bot,” and is planning on replicating himself. We can only hope this means he will never

stop writing, if that is still something humans do in the future.

The Singularity's impact on the economy will be highly disruptive, shifting the focus from deskilling and upskilling to “nonskilling.” This transition is unique compared to previous industrial revolutions, as the emphasis on education has grown alongside labor productivity. Yet Kurzweil does not believe we are in competition with AI. Despite these challenges, employment has grown from 31% to 48% of the population, with per capita GNP increasing by 600% in constant dollars.

Courtesy: ILO

These trends are supported by research, such as Carl Benedikt Frey and Michael Osborne's 2013 [paper](#) and Erik Brynjolfsson and Andrew McAfee in his 2014 [book](#) *The Second Machine Age*, both of which show, to varying degrees, that technology will both eliminate and create jobs. With coding already on the decline, it's essential to adapt to these shifts in the job market and economic landscape. The US had a 45 percent poverty rate in 1870, down to 11.5 percent in 2020. Henrik Ekelund, Founder & Chairman of BTS Group, [wrote](#) in a recent World Economic Forum (WEF) *Agenda* article that concerns today about a “jobless” future will be just as wrong as earlier concerns.

Yet the bigger question is not whether there will be jobs in the future, but rather how to manage the transition. Kurzweil writes: “Although it will be technologically and economically possible for everyone to enjoy a standard of living that is high by today's measures, whether we actually provide this support to everyone who needs it will be a political decision....if we are not careful as a society, toxic politics could interfere with rising living standards.”

Social protection spending in the US has been on the rise, though some argue that current levels are still inadequate. However, as AI continues to

drive down the costs of medicine, food, and housing, it's possible that the percentage of GDP devoted to social safety nets may not need to increase significantly. Nevertheless, Daniel Kahneman cautions that the transition may be marked by conflict and violence.

Grand theories on global net job creation offer little comfort to those living paycheck to paycheck and facing job loss due to AI. The COVID-19 pandemic prompted a basic-income pilot program in the US, with enhanced unemployment benefits, business support, and direct stimulus checks. Just as workers were supported during the pandemic, those who lose jobs due to technological change should be assisted. If progress is for the greater good, the burden of sacrifice should be shared by all, especially those who stand to gain financially, rather than solely by those who lose their jobs.

Kurzweil claims that increasing education has helped us adapt to technological change over the past two centuries. When we merge with non-biological intelligence, reskilling and upskilling will become effortless, as machines can instantly transfer skills to one another through the cloud. Our enhanced neocortex will allow us to download skills instantly, and our intelligence will be digitally backed up. Although “uploading” isn’t expected until the 2040s, Kurzweil suggests keeping written records. In *The Age of Spiritual Machines*, he predicted a 2099 “Destroy-all-copies” movement, enabling individuals to delete their mind file and all backups, raising questions about the control and ownership of digital consciousness.

He foresees an age of abundance where advances in information technology make essential goods and services increasingly affordable. Food and clothing are becoming information technologies, the former reducing violence upon animals. 3D printing is set to revolutionize manufacturing by shifting the paradigm from centralized to decentralized

production. This transformation extends beyond traditional manufacturing and into the realm of biology, enabling the printing of entire organs and even buildings, which could solve homelessness. 3D printing technology is becoming more accessible to non-experts, and is now available at hundreds of UPS locations. In the 2030s, advanced nanomanufacturing will enable the production of nearly anything for mere pennies per pound, thanks to the relentless march of miniaturization.

The main concern for Kurzweil is finding purpose & meaning in a world where many will not have to work if they do not want to. Kurzweil's mentor, Marvin Minsky, [commented](#) that he does not think this will be a problem, as even now folks are easily entertained sitting in a stadium and watching men play football. Such experiences will be enhanced, for, "when we digitally augment our neocortex starting sometime in the 2030s, we will be able to create meaningful expressions that we cannot imagine or understand today." Thanks to AR and VR we will have not just life extension but also "radical life enhancement." In his book *Extend* he argues, "Extending life will also mean vastly *improving* it."

There is also the challenge of trust: "it's not hard to see how exaggerated fears of secret genetic manipulation or government-controlled nanobots could cause people in 2030 or 2050 to reject crucial treatments." What Kurzweil describes as "fundamentalist humanism" will be overcome because demand for therapies will be irresistible.

Kurzweil believes death is a tragedy we rationalize away. He [writes](#): "When we lose that person, we literally lose part of ourselves. This is not just a metaphor—all of the vast pattern recognizers that are filled with the patterns reflecting the person we love suddenly change their nature. Although they can be considered a precious way to keep that person alive within ourselves, the vast neocortical patterns of a lost loved one turn suddenly from triggers of delight to triggers of mourning." He is not

willing to accept it. The promise of the Singularity is to liberate us from our limitations. By extending our lifespan, we can not only live longer but also improve our quality of life, reducing the risk of age-related diseases and enhancing our overall well-being.

Building upon the [ideas presented](#) in his book *Transcend*, we are now entering the second phase of this journey, which involves merging biotechnology with AI. In the 2030s, we will enter a new phase, with nanobots repairing our organs and enabling us to live beyond 120 years. He believes, “We are going to accelerate the extension of our lifespan starting in the 2020s, so if you are in good health and younger than eighty, this will likely happen in your lifetime.” When we begin to utilize all of the earth’s resources, we will find they are a thousand times greater than we need, so overpopulation is not a concern.

The ultimate goal is to put our destiny in our own hands, rather than leaving it to fate, allowing us to live as long as we desire. AI has already demonstrated its potential in improving the speed and quality of COVID-19 vaccines and in computer-aided drug discovery. It also has the potential to target mental health problems at their root cause. As someone who takes many supplements and [expects to be](#) “no older than 40” when the Singularity arrives, Kurzweil embodies the optimism and forward-thinking that characterizes this movement towards a new era of human potential. In *The Singularity is Near*, he writes: “Another error that prognosticators make is to consider the transformations that will result from a single trend in today’s world as if nothing else will change. A good example is the concern that radical life extension will result in overpopulation and the exhaustion of limited material resources to sustain human life, which ignores comparably radical wealth creation from nanotechnology and strong AI.”

Kurzweil’s optimism in his books contrasts with declining reading habits.

While he argues life is improving exponentially, areas like news may not have improved with the shift to digital formats. Kurzweil should address potential downsides, such as shortened attention spans and changing priorities among younger generations. Despite unprecedented access to education, many people choose less intellectually stimulating activities, raising concerns about technology's impact on learning and growth.

The Singularity is Nearer is both a historiography of Kurzweil's work and the field of AI, as well as a significant historical document due to Kurzweil's firsthand experiences. The book should catalyze further exploration of human-machine integration and its implications. Kurzweil's credibility stems from his visionary ideas, once considered outlandish, that have gained traction over time. Although the book covers advanced concepts, its accessibility to the general reader is crucial for fostering a broader societal discussion. It's important for citizens and politicians alike to engage in these conversations and address the ethical, political, legal, and social questions that arise. By doing so, we can proactively manage the development and integration of these transformative technologies.

If we cannot change the future, there is no point in talking about it. Kurzweil is right that the merger between human and machine intelligence is not just inevitable, but already happening. The question, then, is will we have a world akin to Aldous Huxley's *Brave New World*, or one in which we use technology to greatly reduce suffering and increase human potential? A 1903 quote by George Bernard Shaw best sums up Ray Kurzweil, "The reasonable man adapts himself to the world; the unreasonable one persists in trying to adapt the world to himself. Therefore all progress rests on the unreasonable man."

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